Technical Task

Situation

Hello, our future Analytics Engineer. The following exercise is designed to test your ability to derive complex business insights from raw data using **dbt** and **SQL**. We want to see how you approach data modeling problems, make architectural decisions, and handle requirements that aren't fully spelled out.

Resources Provided

You will work with a synthetic SaaS dataset from an imaginary company, which we'll call "Company X". The dataset includes five .csv files:

- accounts.csv (500 rows)
- subscriptions.csv (5,000 rows)
- feature_usage.csv (25,000 rows)
- support_tickets.csv (2,000 rows)
- churn_events.csv (600 rows)

See Appendix 1 and 2 for table schemas and relationships.

Your task

Company X's CFO needs a monthly revenue waterfall to understand MRR movements. Build a **dbt** data model that analysts can use to answer this question.

Step 1: Project Setup

Initialize a **dbt** project and load the five .csv files as source tables. Use any database of your choice.

Step 2: Build Full Data Model

Design and implement a complete **dbt** data model that transforms Company X's raw data into business-ready analytics tables. Your model should follow dimensional modeling best practices with appropriate layering (raw_, staging_, transform_, reporting_).

Step 3: Revenue Waterfall Model

Build a monthly revenue waterfall model that shows how MRR changes from one month to the next. Think about the different ways revenue can increase or decrease, and make sure your model captures all meaningful movements.

Step 4: Data Quality & Documentation

Testing requirements:

- Minimum 5 **dbt** tests total (mix of generic and at least 1 custom test for business logic)

Documentation:

- schema.yml files for your models;
- README.md with modeling decisions and assumptions.

Step 5: Version Control

Host your solution in a public GitHub repository with setup instructions.

(Later) **Step 6: Presentation (~20 mins)**

1. Showcase your **dbt** project and demonstrate running it locally.
2. Walk through your revenue waterfall model and explain the business logic behind it.
3. Discuss limitations of your current implementation and edge cases you've identified.
4. (Important) Propose improvements for a production-ready version, considering:
    - Scalability and performance optimization;
    - Data freshness and incremental strategies;
    - Monitoring and data quality alerting.

Note: Focus on demonstrating your understanding of dimensional modeling, dbt best practices, and analytical thinking. Feel free to use any resources, frameworks, or AI coding tools to complete the exercise.

Good luck!

Appendix 1: Table Schemas

accounts.csv

Column	Type	Description
account_id	ID	Unique customer (primary key)
account_name	string	Fictional company name
industry	categorical	SaaS vertical (e.g., DevTools, EdTech)
country	string	ISO-2 country code
signup_date	date	Account creation date
referral_source	categorical	organic, ads, event, partner, other
plan_tier	categorical	Initial plan (Basic, Pro, Enterprise)
seats	integer	Licensed user count
is_trial	boolean	Currently trialing
churn_flag	boolean	Churned at any point

feature_usage.csv

Column	Type	Description
usage_id	ID	Unique usage event
subscription_id	ID (FK)	Links to subscriptions.subscription_id
usage_date	date	Date of usage
feature_name	categorical	From pool of 40 SaaS features
usage_count	integer	Event frequency
usage_duration_secs	integer	Time spent
error_count	integer	Logged errors
is_beta_feature	boolean	10% flagged as beta

support_tickets.csv

Column	Type	Description
ticket_id	ID	Unique ticket
account_id	ID (FK)	Links to accounts.account_id
submitted_at	datetime	Time opened
closed_at	datetime	Time resolved
resolution_time_hours	float	Duration
priority	categorical	low, medium, high, urgent
first_response_time_minutes	integer	Minutes to first response
satisfaction_score	integer	1–5 (null = no response)
escalation_flag	boolean	True if escalated

churn_events.csv

Column	Type	Description
churn_event_id	ID	Unique churn instance
account_id	ID (FK)	Links to accounts.account_id
churn_date	date	When account left
reason_code	categorical	pricing, support, features, etc.
refund_amount_usd	currency	$0 default, 25% have credit/refund
preceding_upgrade_flag	boolean	Had upgrade within 90 days
preceding_downgrade_flag	boolean	Had downgrade within 90 days
is_reactivation	boolean	10% were previously churned
feedback_text	string	Optional customer comment

subscriptions.csv

Column	Type	Description
subscription_id	ID	Unique subscription (primary key)
account_id	ID (FK)	Links to accounts.account_id
start_date	date	Subscription start
end_date	date	Nullable for active plans
plan_tier	categorical	Plan at time of billing
seats	integer	Licensed seats
mrr_amount	currency	Monthly revenue
arr_amount	currency	Annual revenue
is_trial	boolean	Trial status
upgrade_flag	boolean	Plan upgraded mid-cycle
downgrade_flag	boolean	Plan downgraded mid-cycle
churn_flag	boolean	True if ended
billing_frequency	categorical	monthly or annual
auto_renew_flag	boolean	80% true

Appendix 2: Table Schemas

None

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accounts (PK: account_id)
│
├── subscriptions (FK → accounts.account_id)
│   └── feature_usage (FK → subscriptions.subscription_id)
│
├── support_tickets (FK → accounts.account_id)
└── churn_events (FK → accounts.account_id)
```